Saransh Agrawal

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ML Engineer with production experience in LLM infrastructure, distributed training, and backend systems. Inference optimization via KV Cache Compression (open source @ NVIDIA/kvpress), and published researcher (@ ACL, EACL) - on LLM model alignment, inference and scaling efficiency, mechanistic interpretability.

Education

Texas A&M University *Master of Science in Data Science: (GPA: 3.7/4.0)* *College Station, TX,* ***May 2025***

Courses: Parallel Computing, Database Mgmt., Natural Language Processing, Reinforcement Learning, Data Stream & Algo.

Manipal Institute of Technology *Bachelor's in Electronics and Instrumentation Engineering* ***Aug 2021***

Courses: Data Structures & Algorithms, Microprocessors, Digital Image Processing, Embedded Systems, Digital System Design

Technical Skills

AI/ML Systems: PyTorch, DeepSpeed, Triton, vLLM, llama.cpp, RAG Pipelines, Vector DBs (Qdrant)
Languages: Python, Rust, Go, JavaScript, C++, SQL
Infrastructure: AWS (SageMaker, Bedrock, EKS, Lambda, EC2, S3), Kafka, Docker, Kubernetes, Terraform, HPC
Certificates: AWS Solutions Architect - Professional

Publications and Open Source Contributions

[1] **NVIDIA/kvpress (Open Source):** Contributed a decoding-time KV-cache compression feature reducing cache size by 50% while outperforming baselines by +2% overall and +21% on complex retrieval tasks (RULER). ***Apr 2026***

[2] Ren-Wei Liang, Chin-Ting Hsu, Chan-Hung Yu, **Saransh Agrawal**, Shih-Cheng Huang, Shang-Tse Chen, Kuan-Hao Huang, and Shao-Hua Sun. Adaptive Helpfulness-Harmlessness Alignment with Preference Vectors. In *European Chapter of the Association for Computational Linguistics (EACL) 2026.* ***Mar 2026***

[3] **Saransh Agrawal** and Kuan-Hao Huang. Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) at ACL.* ***July 2025***

Work Experience

AI Engineer | *Saptarishis Software* (saptarishis.net) **Nov 2025 – Present**

- **Intelligent Support Engine:** Engineered a conversational interface to automate product navigation and FAQ resolution; which reduced customer service costs by 35% and accelerated user onboarding.
- **Stateful Reasoning Agent:** Automated a human-advisory service by orchestrating domain knowledge retrieval and state-driven API executions, reducing overhead while maintaining 80% human-expert alignment.

Research Assistant | *FLAIR Lab* (khhuang.me/group.html) **Aug 2024 – May 2025**
NLP Research Lab at Texas A&M University under Prof. Kuan-Hao Huang *College Station, TX*

- **LLM Unlearning:** Developed a novel model training method for removing PII from foundational models. Won 2nd Prize (1B) and highest score (8B) at SemEval-2025; published at SemEval (ACL).
- **Distributed Training Infra:** Achieved 92% scaling efficiency using DeepSpeed ZeRO-3, eliminating network bottlenecks using NCCL; maximizing GPU utilization, reducing experiment runtime and increased research productivity.
- **Inference Optimization:** Developed a vision-language KV-cache merging algorithm (CaM) for LLaVA, reducing VRAM by 21% while maintaining context retrieval accuracy, built for massive image-text contexts.
- **Founding Research Member:** Presented seminars on mechanistic interpretability, and helped establish it as a key research topic for the lab. Collaborated with NTU Taiwan on DPO-based preference vectors (published at EACL).

Software Development Engineer | *Viewzen Labs* (viewzenlabs.com) **Jul 2021 – Jul 2023**
Enterprise MLaaS and data platform startup serving finance, healthcare, and government clients.

- **Multi-tenant MLaaS Platform:** Architected a scalable platform using Go, Python and Kafka to orchestrate experiment tracking and low-latency inference for 50+ enterprise clients.
- **High-Throughput ETL Engine:** Engineered fault-tolerant microservices using a C++/Python FFI, achieving a 15x throughput increase (w.r.t. Node.js) for hierarchical data migrations.
- **MOTHER Project:** Trained and deployed XGBoost models for our client (CDFI), for pregnancy risk prediction with 0.85 F1-score on a highly imbalanced dataset, helping reduce infant mortality rate by 50%.

Machine Learning Engineer Intern | *Centre for Digital Financial Inclusion* (cdfi.in) **Feb 2021 – Jun 2021**
Non-profit developing tech solutions for e-Governance and fintech.

- **Speech-to-SQL Pipeline:** Developed a natural language dashboard creation tool using T5 and schema-guided decoding, achieving 90% SQL validity with sub-4s end-to-end latency, allowing business users to create quick dashboards.
- **ASR Model Fine-tuning:** Reduced Word Error Rate (WER) to 17% by fine-tuning DeepSpeech models on regional Indian accent corpora for e-governance applications.